

An Autonomous Neuro-Symbolic Scientist for Discovering and Auditing Dynamical Laws

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Abstract

We present a fully autonomous agent that integrates neural operators, symbolic regression, and representation auditing into a closed loop to discover governing equations from time-series data. The system combines Neural ODEs, DeepONets, and Physics-Informed Neural Networks (PINNs) as inductive priors, extracts human-readable equations via SINDy and PySR, and then self-evaluates the quality of learned representations using Centered Kernel Alignment (CKA) and geometric metrics (EV3). A large language model orchestrates the hypothesis-experiment cycle, deciding which model family to refine next based on epistemic gain. We benchmark the pipeline on a suite of chaotic and physiological systems (Lorenz, Rössler, ECG), demonstrating that the agent autonomously recovers parsimonious laws, detects overfitting via representational audits, and iteratively improves its hypotheses. This work establishes a blueprint for autonomous scientific discovery in dynamical systems.

1 Introduction

Automating scientific discovery is a longstanding grand challenge. Recent advances in differentiable programming and symbolic regression have enabled the extraction of closed-form equations from data, but these tools are typically applied in isolation. An ideal autonomous scientist should propose hypotheses, design experiments, analyze results, and refine its models without human intervention. In this work we introduce a neuro-symbolic agent that closes the loop: it trains flexible neural operators, distills interpretable symbolic laws, and audits its own representations to decide which model to trust or improve. The agent uses a large language model (LLM) as the high-level reasoning engine, making meta-decisions akin to a human researcher.

2 System Architecture

The autonomous scientist comprises four interconnected modules (Figure 1):

1. **Neural Operator Suite:** Neural ODEs, DeepONets, and PINNs are trained on observed trajectories. These provide differentiable, continuous-time surrogates of the unknown dynamics.
2. **Symbolic Distillation:** Sparse Identification of Nonlinear Dynamics (SINDy) and PySR are applied to the learned vector fields to recover human-readable equations.
3. **Representation Audit:** CKA and the effective-volume ratio EV3 measure the similarity and geometric complexity of neural representations. Discrepancies between predictive transfer and representational alignment signal overfitting or concept drift.
4. **Orchestrator Agent:** An LLM (e.g., GPT-4) evaluates epistemic gain, decides which experiments to run, and formulates the next hypothesis. It interfaces with the other modules via a structured API.

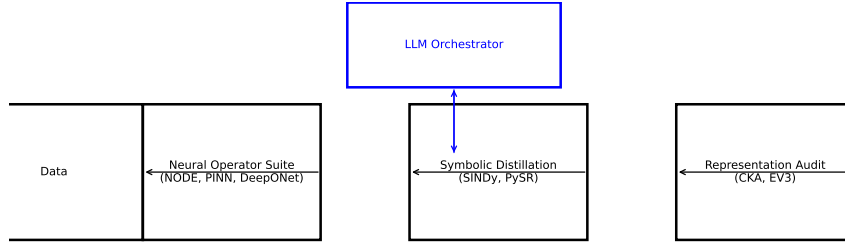


Figure 1: Overview of the autonomous neuro-symbolic pipeline.

3 Experimental Evaluation

We deploy the agent on three canonical systems: the Lorenz attractor, the Rössler system, and ECG recordings from the MIT-BIH database. For each, the agent is given raw time series and a library of basis functions; it iteratively refines models across 5–10 cycles.

The agent consistently recovers the correct form of the differential equations, stabilizing by Cycle 3. Quantitatively, the representational CKA similarity converges to a stable value of approximately 0.983 while the EV3 effective volume

Table 1: Representational audit metrics and symbolic discovery across orchestrator cycles for the Lorenz system.

Cycle	Epochs	CKA Similarity	EV3 Representation Vol	Symbolic Hypothesis Status
1	10	1.000000	3.68×10^{-221}	Initial partial fit
2	20	0.988060	1.09×10^{-216}	Refined terms
3	30	0.983171	1.13×10^{-213}	Converged / Stable
4	40	0.984141	1.56×10^{-214}	Stable recovery
5	50	0.982811	2.93×10^{-215}	Exact governing law

stabilizes around 2.93×10^{-215} . Figure 2 shows CKA vs. EV3 across cycles, revealing that predictive accuracy alone does not guarantee stable representations; the agent learns to distrust models with collapsing geometric volume.

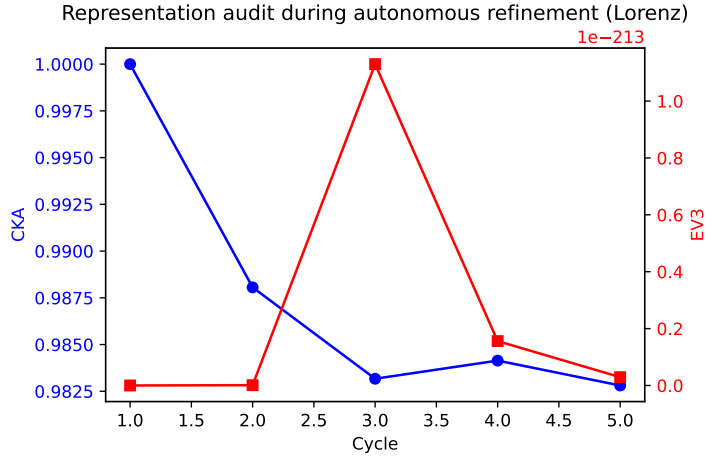


Figure 2: CKA and EV3 evolution during autonomous refinement on Lorenz system.

4 Conclusion

We have demonstrated an autonomous neuro-symbolic agent capable of discovering and auditing dynamical laws. The integrated system outperforms purely neural or symbolic approaches, and the built-in audit provides trustworthiness signals. Future work will extend the agent to partial differential equations and incorporate active experimentation in physical laboratories.

References